Lesson 34: Mixed Strategies, Nash Existence, and the Minimax Theorem

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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4 Mixed Strategies and Mixed Extensions

4.1 Definitions

Definition 4.1 (Mixed Strategy)**.** Let Γ be a finite game with strategy set $S_i = \{s_i^1, \dots, s_i^{m_i}\}$ for player i. A *mixed strategy* for player i is a probability distribution $\sigma_i \in \Delta(S_i)$, where

$$\Delta(S_i) = \left\{ \sigma_i \in \mathbb{R}^{m_i} : \sigma_i(s_i^j) \geq 0 \text{ for all } j, \ \sum_{j=1}^{m_i} \sigma_i(s_i^j) = 1 \right\}$$

is the $(m_i - 1)$-*simplex* over S_i. A *mixed strategy profile* is $\sigma = (\sigma_1, \dots, \sigma_n) \in \Delta := \prod_{i \in N} \Delta(S_i)$.

Definition 4.2 (Support)**.** The *support* of a mixed strategy σ_i is $\text{supp}(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$. A pure strategy s_i is identified with the degenerate distribution placing probability 1 on s_i.

Definition 4.3 (Mixed Extension)**.** The *mixed extension* of a finite game $\Gamma = (N, (S_i), (u_i))$ is the game $\bar{\Gamma} = (N, (\Delta(S_i)), (\bar{u}_i))$ where the expected payoff under independent mixing is

$$\bar{u}_i(\sigma) = \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j) \right) u_i(s) = \mathbb{E}_{s \sim \sigma}[u_i(s)].$$

We drop the bar and write $u_i(\sigma)$ when the context is clear.

Remark 4.1 (Multilinearity). The expected payoff $u_i(\sigma)$ is *multilinear*: it is linear in each σ_i when the other players' strategies are held fixed. Explicitly,

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i) \underbrace{\sum_{s_{-i} \in S_{-i}} \left(\prod_{j \neq i} \sigma_j(s_j) \right) u_i(s_i, s_{-i})}_{=: u_i(s_i, \sigma_{-i})}.$$

The quantity $u_i(s_i, \sigma_{-i})$ is the expected payoff to player i from playing the pure strategy s_i when opponents play σ_{-i}.

4.2 Mixed Nash Equilibrium and the Indifference Principle

Definition 4.4 (Mixed Strategy Nash Equilibrium)**.** A mixed strategy profile σ^* is a *Nash equilibrium* if

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*) \quad \text{for all } \sigma_i \in \Delta(S_i), \text{ for all } i \in N.$$

Theorem 4.1 (Indifference Principle)**.** *Let σ^* be a mixed strategy Nash equilibrium. For each player i, every pure strategy in $\text{supp}(\sigma_i^*)$ yields the same expected payoff against σ_{-i}^*, and this common payoff is at least as large as the expected payoff from any pure strategy outside the support. Formally:*

(a) If $s_i, s_i' \in \text{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) = u_i(s_i', \sigma_{-i}^*)$.

(b) If $s_i \in \text{supp}(\sigma_i^*)$ and $t_i \notin \text{supp}(\sigma_i^*)$, then $u_i(s_i, \sigma_{-i}^*) \geq u_i(t_i, \sigma_{-i}^*)$.

Proof. By multilinearity (Remark 4.1),

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s_i \in S_i} \sigma_i^*(s_i)\, u_i(s_i, \sigma_{-i}^*).$$

Let $v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$ be the equilibrium payoff for player i.

Part (b): Suppose $t_i \in S_i$ satisfies $u_i(t_i, \sigma_{-i}^*) > v_i^*$. Then the pure strategy t_i (viewed as a degenerate mixed strategy) gives payoff $u_i(t_i, \sigma_{-i}^*) > v_i^* = u_i(\sigma_i^*, \sigma_{-i}^*)$, contradicting the NE condition. Hence $u_i(s_i, \sigma_{-i}^*) \leq v_i^*$ for all s_i with equality impossible for all s_i unless we also

have a lower bound. Since v_i^* is a convex combination of the payoffs $u_i(s_i, \sigma_{-i}^*)$ with weights $\sigma_i^*(s_i) \geq 0$ summing to 1, if any payoff in the support were strictly less than v_i^*, then some payoff (in or outside the support) must be strictly greater than v_i^* (to make the weighted average equal v_i^*), which we just showed is impossible. Therefore:

- For all $s_i \in \text{supp}(\sigma_i^*)$: $u_i(s_i, \sigma_{-i}^*) = v_i^*$.
- For all $s_i \notin \text{supp}(\sigma_i^*)$: $u_i(s_i, \sigma_{-i}^*) \leq v_i^*$.

Part (a): Immediate from the first bullet point. □

Example 4.1 (Mixed NE in Battle of the Sexes)**.** In the Battle of the Sexes (Example **??**), let player 1 play O with probability p and player 2 play O with probability q. By the indifference principle, in a fully mixed NE, player 1 must be indifferent:

$$u_1(O, q) = u_1(F, q) \implies 3q + 0(1-q) = 0 \cdot q + 1 \cdot (1-q) \implies 3q = 1 - q \implies q = \tfrac{1}{4}.$$

Similarly, player 2 must be indifferent:

$$u_2(p, O) = u_2(p, F) \implies 1 \cdot p + 0(1-p) = 0 \cdot p + 3(1-p) \implies p = 3 - 3p \implies p = \tfrac{3}{4}.$$

The mixed NE is $\sigma_1 = (\frac{3}{4}, \frac{1}{4})$, $\sigma_2 = (\frac{1}{4}, \frac{3}{4})$, with expected payoffs $(3 \cdot \frac{1}{4}, 1 \cdot \frac{3}{4}) = (\frac{3}{4}, \frac{3}{4})$. Both players fare worse in the mixed NE than in either pure NE.

Mixed Strategies and Stochastic Policies

In RL, a stochastic policy $\pi(a|s)$ assigns probabilities to actions given a state. This is precisely a mixed strategy in the stage game at state s. The indifference principle has a direct analog: in an optimal stochastic policy, every action in the support must yield the same Q-value. If some action had a strictly higher Q-value, the agent should place all probability on it. This is why ϵ-greedy exploration is suboptimal — it assigns probability to actions that may not be in the support of any Nash equilibrium.

5 Nash's Existence Theorem

The central result of this lesson: every finite game has at least one Nash equilibrium in mixed strategies. The proof uses Kakutani's fixed point theorem.

5.1 Kakutani's Fixed Point Theorem

Definition 5.1 (Upper Hemicontinuity)**.** Let $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$. A correspondence (set-valued map) $F : X \rightrightarrows Y$ is *upper hemicontinuous* (uhc) at $x_0 \in X$ if for every open set $V \supseteq F(x_0)$, there exists an open set $U \ni x_0$ such that $F(x) \subseteq V$ for all $x \in U \cap X$.

Equivalently (when Y is compact), F is uhc at x_0 if whenever $x_k \to x_0$ and $y_k \in F(x_k)$ with $y_k \to y_0$, we have $y_0 \in F(x_0)$. This is the *closed graph* characterization.

Theorem 5.1 (Kakutani's Fixed Point Theorem)**.** *Let $K \subseteq \mathbb{R}^n$ be non-empty, compact, and convex. Let $F : K \rightrightarrows K$ be a correspondence such that:*

(i) $F(x)$ is non-empty for every $x \in K$.
(ii) $F(x)$ is convex for every $x \in K$.
(iii) F is upper hemicontinuous.

Then F has a fixed point: there exists $x^* \in K$ with $x^* \in F(x^*)$.

Proof. We provide the standard proof via Brouwer's fixed point theorem.

For each $k \in \mathbb{N}$, let T_k be a triangulation of K with mesh size at most $1/k$ (the maximum diameter of any simplex in the triangulation). For each vertex v of T_k, choose $f_k(v) \in F(v)$ (possible since $F(v)$ is non-empty). Extend f_k to all of K by convex interpolation on each

simplex: if $x = \sum_j \lambda_j v_j$ is the barycentric representation of x in its simplex, set $f_k(x) = \sum_j \lambda_j f_k(v_j)$.

We claim each $f_k : K \to K$ is continuous. Continuity within each simplex follows from the linearity of the interpolation. On shared faces of adjacent simplices, the interpolation agrees because barycentric coordinates are unique on the shared face. Since K is convex and each $f_k(x)$ is a convex combination of points in K, we have $f_k(x) \in K$.

By Brouwer's fixed point theorem (applicable since K is compact and convex, and f_k is continuous), each f_k has a fixed point $x_k \in K$ with $f_k(x_k) = x_k$.

Since K is compact, the sequence (x_k) has a convergent subsequence $x_{k_l} \to x^* \in K$.

We show $x^* \in F(x^*)$. For each l, let σ_l be the simplex of T_{k_l} containing x_{k_l}, with vertices $v_1^l, \ldots, v_p^l$ (where $p \leq n+1$). Then

$$x_{k_l} = f_{k_l}(x_{k_l}) = \sum_{j=1}^{p} \lambda_j^l f_{k_l}(v_j^l), \quad \text{where } x_{k_l} = \sum_{j=1}^{p} \lambda_j^l v_j^l.$$

Since the mesh size tends to 0, $\|v_j^l - x_{k_l}\| \leq 1/k_l \to 0$, so $v_j^l \to x^*$ for each j.

Now, $f_{k_l}(v_j^l) \in F(v_j^l)$. Since K is compact, by passing to a further subsequence (using a diagonal argument over the finitely many vertices), we may assume $f_{k_l}(v_j^l) \to y_j$ for some $y_j \in K$.

By the closed-graph characterization of upper hemicontinuity (Definition 5.1): $v_j^l \to x^*$ and $f_{k_l}(v_j^l) \to y_j$ with $f_{k_l}(v_j^l) \in F(v_j^l)$ implies $y_j \in F(x^*)$.

The barycentric coordinates $\lambda_j^l \in [0,1]$ lie in a compact set, so (passing to a further subsequence) $\lambda_j^l \to \lambda_j^*$ with $\sum_j \lambda_j^* = 1$ and $\lambda_j^* \geq 0$.

Taking the limit in $x_{k_l} = \sum_j \lambda_j^l f_{k_l}(v_j^l)$:

$$x^* = \sum_j \lambda_j^* y_j.$$

Since each $y_j \in F(x^*)$ and $F(x^*)$ is convex, the convex combination $x^* = \sum_j \lambda_j^* y_j \in F(x^*)$. □

5.2 Nash's Theorem

Nash's Existence Theorem

Theorem 5.2 (Nash, 1950). *Every finite normal-form game has at least one mixed strategy Nash equilibrium.*

Proof. Let $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a finite game with $|S_i| = m_i$. Define the set of mixed strategy profiles $\Delta = \prod_{i \in N} \Delta(S_i)$, which is a product of simplices and hence compact and convex in $\mathbb{R}^{m_1 + \cdots + m_n}$.

Step 1: Define the best response correspondence. For each player i, the *mixed best response correspondence* $\mathrm{BR}_i : \Delta_{-i} \rightrightarrows \Delta(S_i)$ is

$$\mathrm{BR}_i(\sigma_{-i}) = \mathrm{argmax}_{\sigma_i \in \Delta(S_i)} u_i(\sigma_i, \sigma_{-i}).$$

Define the joint best response correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$ by

$$\mathrm{BR}(\sigma) = \mathrm{BR}_1(\sigma_{-1}) \times \mathrm{BR}_2(\sigma_{-2}) \times \cdots \times \mathrm{BR}_n(\sigma_{-n}).$$

Step 2: $\mathrm{BR}(\sigma)$ is non-empty. For each i, the function $\sigma_i \mapsto u_i(\sigma_i, \sigma_{-i})$ is continuous (in fact, linear) on the compact set $\Delta(S_i)$. By the extreme value theorem, the maximum is attained, so $\mathrm{BR}_i(\sigma_{-i}) \neq \emptyset$.

Step 3: $\mathrm{BR}(\sigma)$ is convex. Fix i and σ_{-i}. Let $\sigma_i, \sigma_i' \in \mathrm{BR}_i(\sigma_{-i})$ and $\lambda \in [0,1]$. Set $\hat{\sigma}_i = \lambda\sigma_i + (1-\lambda)\sigma_i'$. By linearity of u_i in σ_i (Remark 4.1):

$$u_i(\hat{\sigma}_i, \sigma_{-i}) = \lambda\, u_i(\sigma_i, \sigma_{-i}) + (1-\lambda)\, u_i(\sigma_i', \sigma_{-i}) = \lambda\, v_i^* + (1-\lambda)\, v_i^* = v_i^*,$$

where $v_i^* = \max_{\tau_i} u_i(\tau_i, \sigma_{-i})$. So $\hat{\sigma}_i$ also achieves the maximum, hence $\hat{\sigma}_i \in \mathrm{BR}_i(\sigma_{-i})$. Thus $\mathrm{BR}_i(\sigma_{-i})$ is convex. Since $\mathrm{BR}(\sigma)$ is a product of convex sets, it is convex.

Step 4: BR is upper hemicontinuous. We use the closed-graph characterization. Let $\sigma^k \to \sigma^*$ in Δ and $\tau^k \in \mathrm{BR}(\sigma^k)$ with $\tau^k \to \tau^*$. We must show $\tau^* \in \mathrm{BR}(\sigma^*)$, i.e., $\tau_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$ for each i.

Fix player i. Since $\tau_i^k \in \mathrm{BR}_i(\sigma_{-i}^k)$, for every $\sigma_i \in \Delta(S_i)$:

$$u_i(\tau_i^k, \sigma_{-i}^k) \geq u_i(\sigma_i, \sigma_{-i}^k).$$

Since u_i is continuous (it is multilinear on the compact set Δ), taking $k \to \infty$:

$$u_i(\tau_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*).$$

Since this holds for all σ_i, we have $\tau_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$.

Step 5: Apply Kakutani. The correspondence $\mathrm{BR} : \Delta \rightrightarrows \Delta$ satisfies all hypotheses of Kakutani's theorem (Theorem 5.1): Δ is non-empty, compact, and convex; $\mathrm{BR}(\sigma)$ is non-empty and convex for each σ; and BR is upper hemicontinuous. Therefore, there exists $\sigma^* \in \Delta$ with $\sigma^* \in \mathrm{BR}(\sigma^*)$.

By definition of BR, $\sigma_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$ for all i, which by Proposition **??** (extended to mixed strategies) means σ^* is a Nash equilibrium. □

Remark 5.1 (Alternative Proof via Brouwer). An alternative proof avoids Kakutani by constructing a continuous function whose fixed points are NE. For each player i and pure strategy s_i^j, define the "excess payoff":

$$g_i^j(\sigma) = \max\left\{0,\ u_i(s_i^j, \sigma_{-i}) - u_i(\sigma_i, \sigma_{-i})\right\}.$$

Then define $f : \Delta \to \Delta$ by

$$f_i^j(\sigma) = \frac{\sigma_i(s_i^j) + g_i^j(\sigma)}{1 + \sum_{k=1}^{m_i} g_i^k(\sigma)}.$$

One verifies: (i) f maps Δ to Δ; (ii) f is continuous; (iii) σ^* is a fixed point of f if and only if $g_i^j(\sigma^*) = 0$ for all i, j, which holds if and only if σ^* is a NE. Brouwer's theorem gives a fixed point, hence a NE.

Existence Guarantees in MARL

Nash's theorem guarantees that every finite MARL problem (modeled as a stochastic game with finite state and action spaces) has at least one Nash equilibrium — at least in the stage games. This justifies the goal of Nash-Q learning (Hu & Wellman, 2003), which attempts to compute Nash equilibria at each state. However, the theorem is non-constructive: it guarantees existence but says nothing about efficient computation. Indeed, computing a Nash equilibrium is PPAD-complete (Daskalakis, Goldberg, Papadimitriou, 2009), meaning no polynomial-time algorithm is known.

6 Von Neumann's Minimax Theorem

6.1 Two-Player Zero-Sum Games

Definition 6.1 (Two-Player Zero-Sum Game)**.** A two-player game $(N, (S_1, S_2), (u_1, u_2))$ is *zero-sum* if $u_1(s) + u_2(s) = 0$ for all $s \in S$. It is specified by a single payoff matrix $A \in \mathbb{R}^{m \times n}$ where

$u_1(i,j) = A_{ij}$ and $u_2(i,j) = -A_{ij}$. Player 1 (the "row player") maximizes; player 2 (the "column player") minimizes.

Definition 6.2 (Maximin and Minimax Values)**.** For a zero-sum game with payoff matrix A, define:

$$\underline{v} = \max_{x\in\Delta_m} \min_{y\in\Delta_n} x^\top Ay \quad \text{(maximin value: player 1's guarantee)},$$
$$\overline{v} = \min_{y\in\Delta_n} \max_{x\in\Delta_m} x^\top Ay \quad \text{(minimax value: player 2's guarantee)},$$

where Δ_k denotes the $(k-1)$-simplex in $\mathbb{R}^k$.

Lemma 6.1 (Maximin $\leq$ Minimax)**.** *For any matrix $A \in \mathbb{R}^{m\times n}$, $\underline{v} \leq \overline{v}$.*

Proof. For any fixed $x \in \Delta_m$ and $y \in \Delta_n$:

$$\min_{y'\in\Delta_n} x^\top Ay' \leq x^\top Ay \leq \max_{x'\in\Delta_m} (x')^\top Ay.$$

Taking max over x on the left and min over y on the right:

$$\underline{v} = \max_x \min_{y'} x^\top Ay' \leq \min_y \max_{x'} (x')^\top Ay = \overline{v}. \qquad \square$$

Von Neumann's Minimax Theorem

Theorem 6.2 (Von Neumann, 1928)**.** *For every matrix $A \in \mathbb{R}^{m\times n}$,*

$$\max_{x\in\Delta_m} \min_{y\in\Delta_n} x^\top Ay = \min_{y\in\Delta_n} \max_{x\in\Delta_m} x^\top Ay.$$

The common value $v^* = \underline{v} = \overline{v}$ is called the value *of the game.*

Proof. We give a proof using LP duality. By Lemma 6.1, it suffices to show $\underline{v} \geq \overline{v}$.

Step 1: Reformulate as linear programs. Player 1's maximin problem is:

$$\underline{v} = \max_{x\in\Delta_m} \min_{y\in\Delta_n} x^\top Ay.$$

Since $y \mapsto x^\top Ay$ is linear in y and Δ_n is a simplex, the minimum over y is attained at a vertex e_j (the j-th standard basis vector), giving $\min_y x^\top Ay = \min_{j=1,\dots,n}(A^\top x)_j$.

Thus $\underline{v} = \max_{x\in\Delta_m} \min_j (A^\top x)_j$. Introducing an auxiliary variable v, this is equivalent to the linear program:

$$\begin{aligned} \text{(P)} \quad & \max \ v \\ & \text{s.t.} \ A^\top x \geq v\,\mathbf{1}, \quad x \in \Delta_m, \quad v \in \mathbb{R}, \end{aligned} \tag{1}$$

where $\mathbf{1} = (1,\dots,1)^\top \in \mathbb{R}^n$ and the inequality is componentwise.

Step 2: The dual LP. Similarly, player 2's minimax problem gives:

$$\overline{v} = \min_{y\in\Delta_n} \max_i (Ay)_i.$$

This is equivalent to:

$$\begin{aligned} \text{(D)} \quad & \min \ w \\ & \text{s.t.} \ Ay \leq w\,\mathbf{1}, \quad y \in \Delta_n, \quad w \in \mathbb{R}. \end{aligned} \tag{2}$$

Step 3: LP duality. We show that (P) and (D) are LP duals of each other. Rewrite (P) in standard form. Let $v = \underline{v}$. The constraints of (P) are: $A^\top x - v\,\mathbf{1} \geq 0$, $\mathbf{1}^\top x = 1$, $x \geq 0$. This is a linear program in the variables $(x, v) \in \mathbb{R}^{m+1}$.

The dual of this LP (after careful bookkeeping of the dual variables associated with each primal constraint) yields precisely the LP (D). By strong LP duality (both problems are feasible — take $x = (1/m, \ldots, 1/m)^\top$ and $y = (1/n, \ldots, 1/n)^\top$ — and bounded), the optimal values are equal:

$$\underline{v} = \mathrm{opt}(\mathrm{P}) = \mathrm{opt}(\mathrm{D}) = \overline{v}. \qquad \square$$

Corollary 6.3 (NE in Zero-Sum Games)**.** *In a two-player zero-sum game with payoff matrix A and value v^*:*

(a) (x^*, y^*) is a Nash equilibrium if and only if x^* is a maximin strategy and y^* is a minimax strategy.

(b) All NE yield the same payoff: $u_1 = v^*$, $u_2 = -v^*$.

(c) NE strategies are interchangeable: if (x^*, y^*) and (x^{**}, y^{**}) are both NE, then so are (x^*, y^{**}) and (x^{**}, y^*).

Proof. **(a)** ($\Rightarrow$): Let (x^*, y^*) be a NE. Then x^* is a best response to y^*, so $(x^*)^\top A y^* = \max_x x^\top A y^*$. Similarly, y^* minimizes $x^\top A y$ against x^*, so $(x^*)^\top A y^* = \min_y (x^*)^\top A y$. Therefore:

$$\max_x x^\top A y^* = (x^*)^\top A y^* = \min_y (x^*)^\top A y.$$

But $\min_y (x^*)^\top A y \leq \underline{v} \leq \overline{v} \leq \max_x x^\top A y^*$, and by the minimax theorem $\underline{v} = \overline{v} = v^*$, so all inequalities are equalities. Hence $\min_y (x^*)^\top A y = v^*$ (so x^* achieves the maximin) and $\max_x x^\top A y^* = v^*$ (so y^* achieves the minimax).

($\Leftarrow$): If x^* achieves the maximin and y^* achieves the minimax, then

$$(x^*)^\top A y^* \geq \min_y (x^*)^\top A y = v^* = \max_x x^\top A y^* \geq (x^*)^\top A y^*,$$

so $(x^*)^\top A y^* = v^*$. For any x: $x^\top A y^* \leq \max_{x'} (x')^\top A y^* = v^* = (x^*)^\top A y^*$, so x^* is a best response to y^*. Similarly, y^* is a best response to x^*. Hence (x^*, y^*) is a NE.

(b) Follows from part (a): every NE strategy pair achieves the value v^*.

(c) If x^* and x^{**} are both maximin and y^*, y^{**} both minimax, then by part (a), any combination of a maximin and a minimax strategy forms a NE. $\square$

Minimax in RL

The minimax theorem is the theoretical foundation of:

- **Minimax-Q learning** (Littman, 1994): at each state of a two-player zero-sum stochastic game, the Bellman operator computes $V(s) = \max_x \min_y x^\top Q(s, \cdot, \cdot) y$, which is well-defined by the minimax theorem.
- **AlphaGo/AlphaZero**: the MCTS search implicitly approximates minimax values in the game tree.
- **GANs**: the generator-discriminator game is a continuous zero-sum game $\min_G \max_D V(D, G)$. The minimax theorem (extended to continuous strategy spaces) justifies seeking saddle points.

6.2 Connection to Linear Programming

Proposition 6.4 (Solving Zero-Sum Games via LP)**.** *The maximin strategy x^* and the game value v^* can be computed by solving the LP (1). The minimax strategy y^* is recovered from the dual LP (2). Both LPs have polynomial-time algorithms (e.g., interior point methods), so computing NE in two-player zero-sum games is in P.*

Proof. This follows directly from the reformulation in the proof of Theorem 6.2. The LP (1) has $m+1$ variables and $n+m+1$ constraints (including non-negativity and the simplex constraint), so it can be solved efficiently. □

Zero-Sum vs. General-Sum Complexity

While NE in two-player zero-sum games can be found in polynomial time via LP, computing a Nash equilibrium in a general two-player game is PPAD-complete (Chen, Deng, Teng, 2009; Daskalakis, Goldberg, Papadimitriou, 2009). This complexity-theoretic barrier is a fundamental challenge for MARL: even *finding* the solution concept that algorithms like Nash-Q target is computationally intractable in general.

7 Correlated Equilibrium

Definition 7.1 (Correlated Equilibrium)**.** A *correlated equilibrium* (CE) of a finite game $\Gamma = (N, (S_i), (u_i))$ is a probability distribution π over $S = \prod_{i \in N} S_i$ such that for every player i and every pair of strategies $s_i, s_i' \in S_i$:

$$\sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} \pi(s_i, s_{-i})\, u_i(s_i', s_{-i}).$$

Equivalently, if a mediator draws $s = (s_1, \ldots, s_n)$ from π and privately recommends s_i to player i, no player has an incentive to deviate from the recommendation (assuming the others follow theirs).

Theorem 7.1 (Every Nash Equilibrium Is a Correlated Equilibrium)**.** *If $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed strategy NE, then the product distribution $\pi(s) = \prod_{i \in N} \sigma_i^*(s_i)$ is a correlated equilibrium.*

Proof. Fix player i and strategies $s_i, s_i' \in S_i$. The CE incentive constraint requires:

$$\sum_{s_{-i}} \pi(s_i, s_{-i})\, u_i(s_i, s_{-i}) \geq \sum_{s_{-i}} \pi(s_i, s_{-i})\, u_i(s_i', s_{-i}).$$

Since π is a product distribution, $\pi(s_i, s_{-i}) = \sigma_i^*(s_i) \cdot \prod_{j \neq i} \sigma_j^*(s_j)$. If $\sigma_i^*(s_i) = 0$, both sides are 0 and the inequality holds. If $\sigma_i^*(s_i) > 0$, dividing both sides by $\sigma_i^*(s_i)$ gives:

$$\sum_{s_{-i}} \left(\prod_{j \neq i} \sigma_j^*(s_j) \right) u_i(s_i, s_{-i}) \geq \sum_{s_{-i}} \left(\prod_{j \neq i} \sigma_j^*(s_j) \right) u_i(s_i', s_{-i}),$$

i.e., $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$. Since $s_i \in \operatorname{supp}(\sigma_i^*)$, the indifference principle (Theorem 4.1) gives $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$ for all s_i', which is exactly what we need. □

Proposition 7.2 (CE as a Linear Program)**.** *The set of correlated equilibria of a finite game is a convex polytope, defined by the linear incentive-compatibility constraints in Definition 7.1 together with $\pi(s) \geq 0$ for all s and $\sum_s \pi(s) = 1$. In particular, optimizing any linear objective over the CE set is a linear program.*

Proof. The CE constraints are:

$$\sum_{s_{-i}} \pi(s_i, s_{-i}) \big[u_i(s_i, s_{-i}) - u_i(s_i', s_{-i}) \big] \geq 0 \quad \text{for all } i, s_i, s_i'.$$

These are linear in the variables $\{\pi(s)\}_{s \in S}$. Combined with the simplex constraints $\pi(s) \geq 0$ and $\sum_s \pi(s) = 1$, the feasible set is a polytope (intersection of finitely many half-spaces). Any linear function of π can be optimized over this polytope by solving an LP. □

Example 7.1 (Correlated Equilibrium in Battle of the Sexes)**.** In the Battle of the Sexes (Example **??**), consider the "traffic light" correlated equilibrium: a mediator recommends (O, O) with probability 1/2 and (F, F) with probability 1/2. Under this distribution π:

$$\pi(O, O) = 1/2, \quad \pi(O, F) = 0, \quad \pi(F, O) = 0, \quad \pi(F, F) = 1/2.$$

Check the CE condition for player 1, given recommendation O (so $s_1 = O$):

$$\pi(O, O)\, u_1(O, O) = \tfrac{1}{2} \cdot 3 = \tfrac{3}{2} \geq \pi(O, O)\, u_1(F, O) = \tfrac{1}{2} \cdot 0 = 0. \quad \checkmark$$

Given recommendation F:

$$\pi(F, F)\, u_1(F, F) = \tfrac{1}{2} \cdot 1 = \tfrac{1}{2} \geq \pi(F, F)\, u_1(O, F) = \tfrac{1}{2} \cdot 0 = 0. \quad \checkmark$$

Similarly for player 2. The expected payoffs are $(1/2)(3) + (1/2)(1) = 2$ for player 1 and $(1/2)(1) + (1/2)(3) = 2$ for player 2, which Pareto-dominates the mixed NE payoffs of $(3/4, 3/4)$.

Correlated Equilibrium and MARL

Correlated equilibrium has several advantages over Nash equilibrium for MARL:

- **Computational tractability**: finding a CE is an LP, polynomial in the game size. Finding a NE is PPAD-complete.
- **Natural emergence**: simple no-regret learning dynamics (e.g., regret matching) converge to the set of CE, not NE. This makes CE a more natural prediction for what independent learners will achieve.
- **Better welfare**: CE can achieve higher social welfare than any NE, as the Battle of the Sexes example illustrates.